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APPARATUS FOR GRINDING SURFACE OF STEEL SHEET

Abstract

A steel plate surface polishing device comprises a transfer unit transferring a steel plate, a polishing unit positioned over and coupled to the transfer unit to be rotatable around an axis along a direction perpendicular to the transfer unit, polishing a surface the steel plate through a polishing belt, and adjusting a polishing angle of the steel plate according to rotation, and an adhesion unit coupled to one side of the polishing unit to bring the polishing belt into tight contact with the steel plate.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) [0001] This application is a continuation-in-part (CIP) patent application which claims priority to U.S. patent application Ser. No. 17/802,134, filed on Aug. 24, 2022, which is a national-stage application of International Patent Application No. PCT/KR2021/002300, filed on Feb. 24, 2021, which claims priority to Korean Patent Application No. 10-2020-0025444 filed in the Korean Intellectual Property Office on Feb. 28, 2020, the disclosures of which are incorporated by reference herein in their entireties.

TECHNICAL FIELD

[0002] The disclosure relates to a surface polishing device for steel sheets, and more particularly, to a surface polishing device for steel sheets used for polishing the surface of a steel sheet.

DISCUSSION OF THE RELATED ART

[0003] Forming a scratch pattern, like hairlines, on the surface of a metal plate, e.g., a stainless steel plate, may provide a gorgeous decorative effect while maintaining the inherent luster of the metal and may also achieve anti-fingerprint effects. As such, hairline finishing is widely used as a surface processing method for stainless steel or other metal plates.

[0004] In general, conventional devices for forming hairlines on metal plates form hairlines by rotating or reciprocating a sandpaper-wrapped polishing belt or pad on a steel plate continuously uncoiled or a steel sheet cut to a predetermined length along the feeding or conveying direction of the coiled steel plate or sheet to thereby make scratches. The conventional hairline finishing devices form hairlines along the length direction of the coiled steel plate or the steel sheet cut to a predetermined length and have difficulty in forming hairlines along the horizontal direction of the coiled steel plate or steel sheet.

[0005] Therefore, conventionally, products with a hairline pattern in the length direction of the coil or sheet are mainly supplied, and there is no compact, efficient device for automatically forming a hairline pattern in the horizontal direction of the coiled steel plate or steel sheet. In particular, there is no known efficient, compact device capable of forming hairlines in

[0006] the horizontal direction for both metal plates continuously uncoiled from coiled steel plates and steel sheets cut to a predetermined length. Therefore, conventional hairline-finished products mostly have hairlines in the length direction but there are no products with hairlines in the horizontal direction which may give a novel, aesthetic feeling.

[0007] The conventional hairline forming device typically includes one device unit and one polishing belt or polishing pad. Thus, to form mixed hairlines, which include multiple types of hairlines with different thicknesses, on a single metal plate, multiple hairline forming devices are required, which may cause a complicated device configuration and an increase in space occupation.

SUMMARY

[0008] The disclosure has been devised to address the foregoing issues and aims to polish to have various patterns through a minimal headline device while reducing the size and costs of the device. To achieve the foregoing objectives, according to a preferred embodiment of the disclosure,

[0009] a steel plate surface polishing device comprises a transfer unit transferring a steel plate, a polishing unit positioned over and coupled to the transfer unit to be rotatable around an axis along a direction perpendicular to the transfer unit, polishing a surface the steel plate through a polishing

belt, and adjusting a polishing angle of the steel plate according to rotation, and an adhesion unit coupled to one side of the polishing unit to bring the polishing belt into tight contact with the steel plate.

[0010] Further, a guide rail is coupled to an upper end of the transfer unit to guide a moving roller coupled to an end of the polishing unit and to fix a rotation angle of the polishing unit.

[0011] Further, the guide rail guides the polishing unit to be rotated at an angle ranging from 0 degrees to 45 degrees from a direction perpendicular to a transfer direction of the transfer unit.

[0012] Further, a plurality of the polishing unit are disposed to be symmetrical with each other in a front/rear direction.

[0013] Further, a driving roller driven by a driving motor is coupled to one side of the polishing unit. A horizontal roller positioned in line with the driving roller to be movable in a transverse direction is coupled. A vertical roller is coupled over and between the driving roller and the horizontal roller to be movable in a longitudinal direction. A polishing belt is coupled integrally to the driving roller, the horizontal roller, and the vertical roller. A length of a lowermost end of the polishing belt is adjusted as the horizontal roller and the vertical roller are moved in the transverse direction and the longitudinal direction and the length of the lowermost end is adjusted as the polishing unit is rotated.

[0014] According to the steel plate surface polishing device of the disclosure, it is possible to form various patterns on the surface of a steel plate by adjusting the polishing angle through rotation of the polishing unit rotatably coupled to the transfer unit.

[0015] In particular, various patterns may be formed by adjusting the angle of a single polishing unit, so that it is possible to simplify and downsize the equipment to save costs.

[0016] Further, polishing unit symmetrical to each other may be provided to enable polishing in further diversified patterns.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0017] FIG. 1 is a perspective view illustrating a steel plate surface polishing device according to an embodiment of the disclosure.

[0018] FIG. 2 is a front view of FIG. 1 according to the embodiment of the disclosure.

[0019] FIG. 3 is a front view illustrating an adhesion unit according to an embodiment.

[0020] FIGS. 4 and 5 are views illustrating a polishing unit according to the embodiment of the disclosure.

[0021] FIG. 6 is a front view illustrating a steel plate surface polishing device according to a second embodiment of the disclosure.

[0022] FIG. 7 is a side cross-sectional view illustrating a horizontal roller according to the embodiment of the disclosure.

[0023] FIG. 8 is a front view illustrating the horizontal roller according to the embodiment of the disclosure.

[0024] FIG. 9 is a side cross-sectional view illustrating a vertical roller according to the embodiment of the disclosure.

[0025] FIG. 10 is a front view illustrating the vertical roller according to the embodiment of the disclosure.

[0026] FIG. 11 is a view illustrating operations of the horizontal roller and the vertical roller according to an embodiment of the disclosure.

DETAILED DESCRIPTION

[0027] Advantages and features of the disclosure, and methods for achieving the same may be apparent from the embodiments described below with reference to the accompanying drawings.

[0028] However, the disclosure is not limited to the embodiments disclosed herein, and various changes may be made thereto. The embodiments disclosed herein are provided only to inform one of ordinary skilled in the art of the category of the disclosure. The disclosure is defined only by the appended claims. The same reference numeral denotes the same element throughout the specification.

[0029] Hereinafter, the disclosure is described with reference to the drawings for describing a steel plate surface polishing device according to embodiments of the disclosure.

[0030] FIG. 1 is a perspective view illustrating a steel plate surface polishing device according to an embodiment of the disclosure. FIG. 2 is a front view of FIG. 1 according to an embodiment of the disclosure. FIGS. 3 and 4 are views illustrating a polishing unit according to the embodiment of the disclosure.

[0031] Referring to these drawings, the steel plate surface polishing device according to the present embodiment is able to finish the steel plate surface to have various patterns even with a simple process.

[0032] The steel plate surface polishing device **100** according to an embodiment that may provide such an effect includes a transfer unit **110**, a polishing unit **120**, and an adhesion unit **130**.

[0033] The transfer unit **110** transfers a steel plate **10** and includes a lower body **111** installed on the ground.

[0034] A transfer roller **113** driven by a transfer motor **112** is coupled to an upper end of the lower body **111**.

[0035] At least one or more transfer rollers **113** may be disposed along the transfer direction of the steel plate.

[0036] It is preferable that the transfer roller **113** is positioned higher than the top surface of the lower body **111**.

[0037] This is to prevent the top surface of the lower body **111** from interfering with the polishing unit **120** when the steel plate **10** transferred by the transfer roller **113** is polished by the polishing unit **120** to be described below.

[0038] A guide rail **114** which is curved from one side, as a reference point, of the transfer roller **113** to the other side is coupled to the top end of the lower body **111**.

[0039] In this case, the guide rail **114** is curved by about 45 degrees forward along the direction perpendicular to the transfer direction of the transfer roller **113**.

[0040] The polishing unit **120** is installed over the transfer unit **110** to polish the surface of the steel plate **10** transferred by the transfer roller **113**. The polishing unit **120** includes a rotation shaft **121** coupled to a reference point of the lower body **111** and an upper body **122** coupled to a side of the rotation shaft **121**.

[0041] Specifically, the rotation shaft **121** is rotatably coupled to the reference point of the lower body **111**.

[0042] The upper body **122** has one side rotatably and integrally coupled to the rotation shaft **121** and the other side extending to the other side of the transfer roller **113**.

[0043] A moving roller **123** guided along the guide rail **114** is coupled to a lower portion of the other end of the upper body **122**.

[0044] The upper body **122** is rotated by about 45 degrees from the direction perpendicular to the transfer direction of the transfer roller **113**, along the guide rail **114**, by the moving roller **123**.

[0045] The moving roller **123** has a brake (not shown) and may fix the rotation angle of the upper body **122**.

[0046] A driving motor M is coupled to one side of the rear of the upper body **122**, and a sub roller R connected with the driving motor M by a driving belt B is coupled to a lower end of the one side of the rear.

[0047] A driving roller **125** connected with the sub roller R to be integrally driven is coupled to a lower end of the one side of the front of the upper body **122**.

[0048] A horizontal roller **126** disposed in line with the driving roller **125** is rotatably coupled to a lower end of the other side of the front of the upper body **122**.

[0049] A vertical roller **127** disposed over and between the driving roller **125** and the horizontal roller **126**, is rotatably coupled to the front of the upper body **122**.

[0050] In this case, the driving roller **125**, the horizontal roller **126**, and the vertical roller **127** are disposed in a substantially triangular shape.

[0051] One polishing belt **128** is coupled to integrally surround the driving roller **125**, the horizontal roller **126**, and the vertical roller **127**.

[0052] The polishing belt **128** is rotated together with the horizontal roller **126** and the vertical roller **127** to polish the steel plate **10** according to the driving of the driving roller **125**.

[0053] In this case, a pressing roller **124** is rotatably coupled to the frontmost lower end of the upper body **122** to press the steel plate **10**.

[0054] The pressing roller **124** presses and fixes the steel plate **10** to prevent the steel plate **10** from being shaken by the polishing belt **128** while being polished.

[0055] Thus, the polishing unit **120** may form hairlines in a diagonal direction between 0 degrees and 45 degrees, as well as hairlines in the horizontal direction, on the steel plate **10** through rotation.

[0056] Further, the driving roller **125**, the horizontal roller **126**, and the vertical roller **127** may be formed in duplicate. The polishing belt **128** may also be coupled in duplicate.

[0057] Further, the polishing unit **120** may be installed at the front and rear, respectively, of the transfer unit **110** and be symmetrical to each other to polish the steel plate **10** substantially in an “x” shape when rotated at 45 degrees therebetween.

[0058] The adhesion unit **130** is to bring the polishing belt **128** into close contact with the steel plate **10**, and includes a first driver **131** coupled to one side of the upper body **122** and a second driver **135** coupled to one side of the lower body **111**.

[0059] The first driver **131** includes a first driving motor **131a** coupled to one side of the upper body **122**, a first driving shaft **131b** driven by the first driving motor **131a**, and a first rotating roller **131c** coupled to the first driving shaft **131b**. The first rotating roller **131c** is disposed at a position adjacent to the driving roller **125**.

[0060] The first driver **131** includes a first interlocking roller **131d** spaced apart from the first rotating roller **131c** by a predetermined distance and disposed at a position adjacent to the horizontal roller **126** and a first driving belt **131e** coupled to the first rotating roller **131c** and the first interlocking roller **131d**.

[0061] The first reciprocating member **132** is coupled to the first driving belt **131e** so as to be movable together according to driving, and the first reciprocating member **132** extends to be positioned between the driving roller **125** and the horizontal roller **126**. A first contact roller **133** is coupled to an end of the first reciprocating member **132** so as to tightly contact the upper surface of the lower portion of the polishing belt between the driving roller **125** and the horizontal roller **126**.

[0062] A first tread **133a** is formed on the circumference of the first contact roller **133** along the transverse direction, which is the moving direction of the first reciprocating member **132**.

[0063] The adhesion unit **130** includes a first height adjuster **134** for adjusting the height of the first driver **131** coupled to the upper body **122**.

[0064] The first height adjuster **134** may adjust the height of the first contact roller **133** by adjusting the height of the first driver **131**.

[0065] The second driver **135** includes a second driving motor **135a** coupled to one side of the lower body **111**, a second driving shaft **135b** driven by the second driving motor **135a**, and a second rotating roller **135c** coupled to the second driving shaft **135b**. The second rotating roller **135c** is disposed at a position facing the first rotating roller **131c**.

[0066] The second driver **135** includes a second interlocking roller **135d** spaced apart from the second rotating roller **135c** by a predetermined distance and disposed at a position facing the first

interlocking roller **131d**, and a second driving belt **135e** coupled to the second rotating roller **135c** and the second interlocking roller **135d**.

[0067] The second reciprocating member **136** is coupled to the second driving belt **135e** so as to be movable together according to driving, and the second reciprocating member **136** extends symmetrically with the first reciprocating member **132**.

[0068] A second contact roller **137** is coupled to an end of the second reciprocating member **136** so as to tightly contact the lower surface of the steel plate **10**.

[0069] A second tread **137a** disposed in the longitudinal direction, which is the direction in which the steel plate **10** is transferred, is formed on the circumference of the second adhesion roller **137**. The second tread **137a** is disposed to be perpendicular to the first tread **133a**.

[0070] The adhesion unit **130** includes a second height adjuster **138** for adjusting the height of the second driver **135**, coupled to the lower body **111**.

[0071] The second height adjuster **138** may adjust the height of the second contact roller **137** by adjusting the height of the second driver **135**.

[0072] Meanwhile, the first driver **131** and the second driver **135** rotate repeatedly in one direction and the other direction so that the first reciprocating member **132** and the second reciprocating member **136** reciprocate between the two opposite sides.

[0073] FIG. **6** is a front view illustrating a steel plate surface polishing device according to an embodiment of the disclosure. FIG. **7** is a side cross-sectional view illustrating a horizontal roller according to the embodiment of the disclosure. FIG. **8** is a front view illustrating the horizontal roller according to an embodiment of the disclosure. FIG. **9** is a side cross-sectional view illustrating a vertical roller according to an embodiment of the disclosure. FIG. **10** is a front view illustrating the vertical roller according to an embodiment of the disclosure. FIG. **11** is a view illustrating operations of the horizontal roller and the vertical roller according to an embodiment of the disclosure.

[0074] The steel plate surface polishing device **200** according to the present embodiment includes a transfer unit **110** and a polishing unit **220**.

[0075] The transfer unit **110** transfers a steel plate **10** and includes a lower body **111** installed on the ground.

[0076] A transfer roller **113** driven by a transfer motor **112** is coupled to an upper center of the lower body **111**.

[0077] The polishing unit **220** is installed over the transfer unit **110** to polish the surface of the steel plate **10** transferred by the transfer roller **113**. The polishing unit **120** includes a rotation shaft **221** coupled to a reference point of the lower body **111** and an upper body **222** coupled to a side of the rotation shaft **221**.

[0078] A driving motor **M** is coupled to one side of the rear of the upper body **222**, and a sub roller **R** connected with the driving motor **M** by a driving belt **B** is coupled to a lower end of the one side of the rear.

[0079] A driving roller **225** connected with the sub roller **R** to be integrally driven is coupled to a lower end of the one side of the front of the upper body **222**.

[0080] A horizontal roller **226** disposed in line with the driving roller **225** is coupled to a lower end of the other side of the front of the upper body **222**, rotatably and movably in the traverse direction.

[0081] A vertical roller **227** disposed over and between the driving roller **225** and the horizontal roller **226**, is coupled to the front of the upper body **222**, rotatably and movably in the longitudinal direction.

[0082] In this case, the driving roller **225**, the horizontal roller **226**, and the vertical roller **227** are disposed in a substantially triangular shape.

[0083] One abrasive belt **228** is coupled to integrally surround the driving roller **225**, the horizontal roller **226**, and the vertical roller **227**.

[0084] The polishing belt **228** is rotated together with the horizontal roller **226** and the vertical

roller **227** to polish the steel plate **10** according to the driving of the driving roller **225**.

[0085] In this case, a pressing roller **224** is rotatably coupled to the frontmost lower end of the upper body **222** to press the steel plate **10**.

[0086] The pressing roller **224** presses and fixes the steel plate **10** to prevent the steel plate **10** from being shaken by the polishing belt **228** while being polished.

[0087] Meanwhile, the horizontal roller **226** and the vertical roller **227** may be movably coupled to the upper body **222** and be fixed in a moved position and, as moved, be able to adjust the lowermost length of the polishing belt **228**.

[0088] Specifically, a first long hole **222a** is formed, in the transverse direction, in the lower end of the other side of the front of the upper body **222**, and a first moving member **226a** is movably coupled to the first long hole **222a**.

[0089] The horizontal roller **226** is rotatably coupled to the front of the first moving member **226a**.

[0090] Further, first guide wings **226b** extending in the upper and lower directions of the first long hole **222a** are formed in the first moving member **226a** in the front/rear direction of the upper body **222**.

[0091] Here, a first fixing bolt **226c** for fixing the position of the first moving member **226a** is fastened to the first guide wing **226b** positioned in front of the upper body **222** among the first guide wings **226b**.

[0092] In this case, an end of the first fixing bolt **226c** presses the front of the upper body **222** to fix the first moving member **226a** according to the degree of fastening.

[0093] A second long hole **222b** is formed, in the longitudinal direction, in the upper center of the front of the upper body **222**, and a second moving member **227a** is movably coupled to the second long hole **222b**.

[0094] The vertical roller **227** is rotatably coupled to the front of the second moving member **227a**.

[0095] Further, second guide wings **227b** extending in two lateral directions of the second long hole **222b** are formed in the second moving member **227a** in the front/rear direction of the upper body **222**.

[0096] Here, a second fixing bolt **227c** for fixing the position of the second moving member **227a** is fastened to the second guide wing **227b** positioned in front of the upper body **222** among the second guide wings **227b**.

[0097] In this case, an end of the second fixing bolt **227c** presses the front of the upper body **222** to fix the second moving member **227a** according to the degree of fastening.

[0098] Accordingly, the positions of the horizontal roller **226** and the vertical roller **227** may be adjusted.

[0099] Meanwhile, the length of the lowermost end of the polishing belt **228** is adjusted according to the positions of the horizontal roller **226** and the vertical roller **227**.

[0100] Specifically, the length of the lowermost end of the polishing belt **228** is reduced when the horizontal roller **226** moves in one lateral direction and increased when the horizontal roller **226** moves in the opposite lateral direction.

[0101] In this case, the vertical roller **227** is moved upward, when the horizontal roller **226** moves in the one lateral direction, and moved downward when moving in the opposite direction so as to constantly maintain the tension of the polishing belt **228** according to the upper/lower movements.

[0102] Here, since the contact surface of the polishing belt **228** with the steel plate **10** varies depending on the rotation angle of the polishing unit, the length of the lowermost end thereof is preferably adjusted to correspond thereto.

[0103] For example, when the rotation angle of the polishing unit **220** is 45 degrees, the polishing unit **220** is positioned in an oblique direction with respect to the transfer roller **113**, so that the area of contact with the transfer roller **113** is larger than that when the polishing unit **220** is at a rotation angle of 0 degrees.

[0104] Therefore, it is preferable to adjust the length of the lowermost end of the polishing belt **228**

according to the rotation angle of the polishing unit **220** so that the area in contact with the transfer roller **113** is the same.

[0105] Here, when the polishing unit **220** is positioned in a direction perpendicular to the transfer roller **113**, the rotation angle is 0 degrees, and the area of contact with the polishing belt **228** is relatively short.

[0106] Accordingly, the polishing belt **228** provides an optimal polishing area by adjusting the length of the lowermost end through the horizontal roller **226** and the vertical roller **227** according to the rotation angle of the polishing unit **220**.

[0107] By such a configuration, the polishing unit **220** may form hairlines in a diagonal direction between 0 degrees and 45 degrees, as well as hairlines in the horizontal direction, on the steel plate **10** according to the rotation.

[0108] According to the steel plate surface polishing device of the disclosure, it is possible to form various patterns on the surface of a steel plate by adjusting the polishing angle through rotation of the polishing unit rotatably coupled to the transfer unit.

[0109] In particular, various patterns may be formed by adjusting the angle of a single polishing unit, so that it is possible to simplify and downsize the equipment to save costs.

[0110] Further, polishing units symmetrical to each other may be provided to enable polishing in further diversified patterns.

[0111] It will be appreciated by one of ordinary skill in the art that the disclosure may be implemented in other various specific forms without changing the essence or technical spirit of the disclosure. Thus, it should be noted that the above-described embodiments are provided as examples and should not be interpreted as limiting. It should be noted that the scope of the disclosure is defined by the appended claims rather than the described description of the embodiments and include all modifications or changes made to the claims or equivalents of the claims.

Claims

1. A steel plate surface polishing device, comprising: a transfer unit transferring a steel plate; a polishing unit positioned over and coupled to the transfer unit to be rotatable around an axis along a direction perpendicular to the transfer unit, polishing a surface the steel plate through a polishing belt, and adjusting a polishing angle of the steel plate according to rotation; and an adhesion unit coupled to one side of the polishing unit to bring the polishing belt into tight contact with the steel plate.
2. The steel plate surface polishing device of claim 1, wherein a guide rail is coupled to an upper end of the transfer unit to guide a moving roller coupled to an end of the polishing unit and to fix a rotation angle of the polishing unit.
3. The steel plate surface polishing device of claim 2, wherein the guide rail guides the polishing unit to be rotated at an angle ranging from 0 degrees to 45 degrees from a direction perpendicular to a transfer direction of the transfer unit.
4. The steel plate surface polishing device of claim 3, wherein a plurality of the polishing unit are disposed to be symmetrical with each other in a front/rear direction.
5. The steel plate surface polishing device of claim 3, wherein a driving roller driven by a driving motor is coupled to one side of the polishing unit, wherein a horizontal roller positioned in line with the driving roller to be movable in a transverse direction is coupled, wherein a vertical roller is coupled over and between the driving roller and the horizontal roller to be movable in a longitudinal direction, wherein a polishing belt is coupled integrally to the driving roller, the horizontal roller, and the vertical roller, wherein a length of a lowermost end of the polishing belt is adjusted as the horizontal roller and the vertical roller are moved in the transverse direction and the

longitudinal direction and the length of the lowermost end is adjusted as the polishing unit is rotated.
